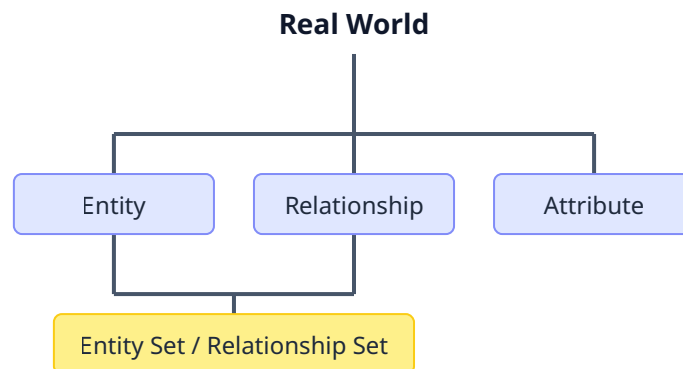


# ER MODEL NOTES

Everything in an ER model revolves around four concepts.



**Think of it like this:**

**Entity** → Object

**Attribute** → Property of the object

**Relationship** → Connection between objects

**Entity Set** → Collection of similar objects .

## What is an Entity?

### Definition

An Entity is a distinguishable real-world object that has an independent existence and about which information is stored in the database. an entity must be uniquely identifiable

## Characteristics of an Entity

A good entity:

- Exists independently
- Has attributes
- Can be uniquely identified
- Represents a real-world object.

## What is an Entity Set?

### Definition

An Entity Set is a collection of entities of the same type that share the same attributes.

### Student Entity Set

#### Student

Contains:

- Aadya
- Arpit

Every one of them has:

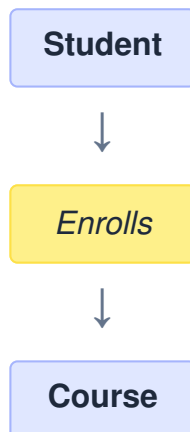
- Roll Number
- Name
- CGPA

Same attributes.

## What is a Relationship?

### Definition

A Relationship is an association between two or more entities.



Student and Course are entities.  
"Enrolls" is the relationship.

## Relationship Set

### Definition

A Relationship Set is the collection of all relationships of the same type between entity sets.

### Example

#### Student

Student  
Aadya  
Arpit

#### Course

Course  
VLSI  
MWE

**Relationships:**

- Aadya  $\rightarrow$  VLSI
- Aadya  $\rightarrow$  MWE
- Arpit  $\rightarrow$  VLSI

All of these "Enrolls" relationships together form the Relationship Set

## ER View vs Relational View

| ER Model                                                                           | Relational Database                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Student</b></p> <p>↓</p> <p><i>Enrolls</i></p> <p>↓</p> <p><b>Course</b></p> | <p><b>Students</b></p> <p>  StudentID   Name  </p> <p><b>Courses</b></p> <p>  CourseID   Name  </p> <p><b>Enrollments</b></p> <p>  StudentID   CourseID   Grade  </p> <p>, grade here is a relation attribute , attribute of relation itself , grade does not belong to student or course , it belongs to enrollement , itslef</p> |

| ER Model                                                                           | Relational Model                                                                                                               |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Student</b></p> <p>↓</p> <p><i>Enrolls</i></p> <p>↓</p> <p><b>Course</b></p> | <p><b>Enrollment</b></p> <p>StudentID (FK)</p> <p>CourseID (FK)</p> <p><i>The Foreign Keys implement the relationship.</i></p> |

An entity is a conceptual real-world object. A table is its implementation in a relational database.

A one-to-many relationship is usually implemented using a foreign key.

A many-to-many relationship is implemented using a separate junction (bridge) table containing foreign keys.

A Relationship is the conceptual association between entities in the ER model. A Foreign Key is the database mechanism used to implement that relationship in a relational schema.